

BSFL Commercial Launch — Financial Projections

Site 1 · Lockyer Valley QLD · May–December 2026 · Realistic scenario

30 hrs/wk owner 22 physical, 8 admin	\$35/hr casual hired month 7+ (8 hrs/wk)	Credit card buffer ~\$3k avg, 0.05%/day	Self-harvesting design reduces labour 40%	5 kg waste → 1 kg fresh larvae (mid range)	12-day avg cycle Lockyer Valley warmth
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Month	Output kg/wk	Live Revenue	Dried Revenue	Frass Revenue	Chitin Revenue	Stall Revenue	Total Revenue	Total Costs	Net Profit	Margin %
May	15	\$1,546	\$195	\$572	—	—	\$2,312	\$718	\$1,595	69%
Jun	22	\$2,134	\$381	\$838	—	—	\$3,353	\$828	\$2,525	75%
Jul	28	\$2,546	\$606	\$1,067	—	—	\$4,219	\$973	\$3,247	77%
Aug	35	\$2,758	\$1,061	\$1,334	—	\$1,200	\$6,353	\$1,289	\$5,064	80%
Sep	45	\$3,273	\$1,559	\$1,715	\$281	\$1,400	\$8,228	\$1,484	\$6,744	82%
Oct	55	\$3,668	\$2,143	\$2,096	\$343	\$1,400	\$9,650	\$1,639	\$8,011	83%
Nov	58	\$3,868	\$2,260	\$2,210	\$362	\$1,500	\$10,199	\$2,817	\$7,383	72%
Dec	60	\$4,001	\$2,338	\$2,286	\$374	\$1,600	\$10,599	\$2,859	\$7,741	73%
TOTAL							\$54,913	\$12,605	\$42,308	77%

Price & output assumptions

Product	Price Used	Basis	Volume Ramp
Live BSFL (pet/poultry)	\$28/kg blended	\$22/kg wholesale to stores; \$35/kg direct. Blended 60/40.	15 kg/wk (May) → 40 kg/wk (Dec)
Dried BSFL	\$80/kg	Online and market. Conservative vs \$100/kg retail ceiling.	0.9 kg/wk (May) → 8 kg/wk (Dec) — grows as market builds
Frass fertiliser	\$11/kg blended	\$14/kg retail bags; \$9/kg farm bulk contract. Ratio 50/50 by month 4.	~12 kg/wk (May) → ~56 kg/wk (Dec)
Chitin powder (agric. grade)	\$30/kg	Conservative agric. grade. 6% of dried frass weight. Starts month 5.	~0.6 kg/wk (Sep) → ~2.5 kg/wk (Dec)
Tipping fees	\$55/tonne waste	Waste suppliers pay to divert from landfill (\$75/t landfill levy avoided). Month 7+.	~1.5 t/wk from Nov at full output
Market stall	\$600–800/day	Fortnightly. Live + dried + frass + chitin. Grows as reputation builds.	2 days/month from Aug

Cost assumptions

Cost Item	Rate	Notes
Packaging & labels	\$1.50/kg product	Kraft bags, labels, tape. Applied to all product kg — live, dried, frass, chitin.
Feed supplement (wheat bran blend)	\$0.50/kg larvae	15% bran at \$0.40/kg blended into substrate for nutritional consistency. Pure food waste is unreliable as sole substrate. Applied to larvae kg produced.

Consumables (HCl, NaOH, biosecurity reagents)	\$0.50/kg product	Chitin extraction chemicals, disinfectants, cleaning supplies per kg product processed.
Power (dehydrator, climate control)	\$130/month	Increases slightly vs early model as dehydrator runs harder at scale. Lockyer summer reduces heating cost.
Equipment maintenance / depreciation	\$45/month	Dehydrator elements, trays, screens, containers — realistic wear on \$60 second-hand unit running continuously.
Biosecurity & hygiene	\$50/month	Bleach, disinfectant, PPE, fly screens, rodent control. A working farm goes through this constantly.
Insurance (product liability)	\$15/month	\$180/year recurring. Previously included only as startup cost — now correctly modelled as ongoing.
Accounting / BAS compliance	\$30/month	Quarterly BAS lodgement support, Wave Accounting, occasional accountant query.
Phone (business portion)	\$60/month	Order management, Facebook group sales, microscope photos, waste supplier coordination.
Bank & card processing fees	\$30 + 1.5% stall card	Business account flat fee + card processing at market stall (EFTPOS/Square). ~\$30–50/month.
Casual labour (month 7+)	\$35/hr × 8 hrs/wk	\$1,120/month from month 7. Enables output ramp to 58–60 kg/week.
Credit card interest	~\$15/month	~\$3k avg revolving balance. Used for timing float on bulk orders — target cleared monthly.
Transport (all components)	\$91–315/month	Waste collection runs (2–3×/wk, 40km avg return), wholesale delivery (60km/wk from month 3), stall travel (100km return × 2/month from month 4), parcel post for online dried/chitin (~\$12/parcel × 8/month). Live larvae local pickup only.
Tipping fees income	\$0 in Year 1	Waste suppliers provide waste FREE in Year 1 (saves them contractor cost). Formal tipping fee agreements require 6–12 months operating history and documented waste management credentials — a Year 2 revenue item.
Market stall fees	\$55/day	Lockyer Valley and Toowoomba farmers markets from month 4.
Owner wage	\$0 (drawings)	All net profit taken as owner drawings. Not a cost — net profit IS your income.

77%
Avg gross
margin

Mo
First pro

- Previously missing costs now included: feed supplement (\$0.50/kg larvae), equipment maintenance (\$45/mo), biosecurity (\$50/mo), insurance (\$15/mo), accounting (\$30/mo), phone (\$60/mo), bank/card fees (~\$35/mo). These add ~\$235–290/month to costs.
- Yield buffer applied: 10% reduction in larvae output vs prior model to reflect real-world substrate inconsistency. Pure food waste alone is nutritionally variable — bran supplement stabilises conversion.
- Tipping fees removed from Year 1 entirely. Waste is received free (saving suppliers contractor cost). Formal tipping fee contracts require operating history and waste management credentials — realistic from Year 2.
- Transport fully modelled at \$1,763 across 8 months: waste collection runs, wholesale delivery, stall travel, parcel post.
- Live larvae not posted interstate — local pickup and direct delivery only. Express post at \$12–16/pack destroys margin on \$22–28 live product.
- Net profit of \$42,308 on \$54,913 revenue at 77% margin is realistic for a physical-goods business with strong IP (biological knowledge). Margin improves in Year 2 as fixed costs spread over higher volume.

Prepared April 2026 · Based on mid-range BSFL bioconversion yields (Rehman et al., 2022), current Australian market pricing, and Queensland labour rates. Projections are estimates — actual results depend on waste supply confirmation, customer acquisition timing, and rearing performance.